Python Basics Notes

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1 Variables

- Variables is a name in python refers to the value stored in memory.
- Value is stored once and used multiple times.
- Easy to change a value in place instead of changing everywhere.
- Variables are used to store user input.
- Making decisions.

Syntax

```
variable_name = value
```

2 Basic Data Types

Data type tells us which type of value is stored in a variable.

a. Integer

- It refers to the number without decimal point.
- Integer is used when we have to represent countable things (Age, Roll_no).

b. Float

- Float is a number with decimal point.
- Some values are not in integers like height, weight; then we use float.

c. Boolean

- Boolean has only 2 values: True and False.
- It is mostly used in decision making.

d. String

- String is a sequence of characters.
- String is used when text values are declared to variables (e.g., Name, email).

3 Arithmetic Operators

- These are symbols which perform mathematical operation on numbers.
- These operators are used to perform calculation.
- Arithmetic operators are addition (`+`), subtraction (`-`), multiplication (`*`), division (`/`), floor division (`//`), exponent (`**`), modulus (`%`).

Syntax

```
operand1 operator operand2
```

4 Comparison Operators

- These are the operators that compare two values and return the result in the form of Boolean values.
- They are mainly used in decision making.
- Comparison operators are Equal to (`==`), Not equal to (`!=`), greater than (`>`), less than (`<`), greater than equal to (`>=`), less than equal to (`<=`).

Syntax

```
operand1 operator operand2
```

5 Assignment Operators

- These are the operators that assign a value to a variable.
- The assignment operators are Assignment (`=`), Add and assign (`+=`), Subtract and assign (`-=`), Multiply and assign (`*=`), Divide and assign (`/=`), Floor divide and assign (`//=`), Exponent and assign (`**=`), Modulus and assign (`%=`).

Syntax

```
operand1 operator operand2
```

6 Type Conversion

- This is the process of converting a value from one data type to another.
- We use type conversion because different data types cannot always be used together.
- For type conversion python provides built-in functions: for Integer it is `int()`, for String it is `str()`, for Float it is `float()`, and for Boolean it is `bool()`.

7 Input and Output

- Input and output functions are used to accept data from user and display it.
- `input()` – accepts data from user.
- `print()` – displays data to user.

8 Strings

- String is a sequence of characters enclosed in quotes.
- String can contain letters, symbols, numbers.

String Creation

- String creation is the process of creating a string value and storing it in a variable so later we can manipulate the text data in a program.
- It is created by placing characters in single quotes (`' '`), double quotes (`" "`), and triple quotes (`""" """` or `''' '''`).

Indexing

- In this we access the individual element of a string using their position.
- It is used when we have to work with a specific character in a string.

Syntax

```
string[index]
```

Slicing

- In this a substring is extracted from a string using an index range.
- Index can access a single character but slicing can access multiple characters.

Syntax

```
string[start_index:end_index]
```

String Operations

- String concatenation (`+`) – joins 2 or more strings.
- String repetition (`*`) – repeats a string multiple times. Mostly used in creating patterns.

There are also some string functions to convert string to lowercase (`lower()`), uppercase (`upper()`), and finding string length (`len()`).

9 Lists

- A list is an ordered collection of multiple values.
- A list can store strings, numbers, and different data types together.

Creating a List

List is created using square brackets `[]` or the `list()` function.

Indexing

- It is similar to string indexing, accessing the element based on their position.
- Without indexing we cannot access a specific value from a list.

Syntax

```
list_name[index]
```

Slicing

- Here the sublist from a list is accessed based on their position using a range.
- Indexing accesses one element while slicing accesses multiple elements.

Updating Elements

- Updating elements in a list means changing or replacing the existing value of an element with a new value.
- Because lists are mutable, we can modify list elements after creating a list.

Nested Lists

- Nested list means a list within a list, i.e., the list has elements in the form of lists.
- We need nested lists mainly when the data is in tabular form.

Syntax

```
list = [[list1], [list2], [list3]]
```

Basic List Methods

a. **append()**

- Method used to add a new element at the end of the list.
- Used when we want to add new data in an existing list without creating a new list.

Syntax

```
list_name.append(value)
```

b. **insert()**

- When we want to add data at a particular location we use `insert()`.

Syntax

```
list_name.insert(index, value)
```

c. **remove()**

- Used to delete the first occurrence of a specific value from a list.

Syntax

```
list_name.remove(value)
```

d. **pop()**

- Used to remove and return an element from a list using its index; if index is not given then it removes the last element.

Syntax

```
list.pop(index)
```

e. **sort()**

- Used to arrange elements of a list in ascending or descending order.

Syntax

```
list.sort()
```

f. **reverse()**

- Used to reverse the order of elements in a list.

Syntax

```
list.reverse()
```

g. **len()**

- Used to find the number of elements in a list.

Syntax

```
len(list)
```

10 Mini Experiments

GitHub link: https://github.com/Prapti-gagare/Python